Lab 1: Regression Models

DA380A: Machine Learning

The purpose of this lab is to practice the complete workflow for a regression problem: exploring data, preparing it for machine learning, building regression models, evaluating their performance, and comparing the results.

Provided datasets

Choose one of the datasets provided for Lab 1:

No.	Provided dataset
1	dataset1-VideoReviews
2	dataset2-SalesTransactions2
3	dataset3-AgriculturalProduction
4	dataset4-SalesTransactions1
5	dataset5-AirTravel

After selecting a dataset, identify a suitable numerical target variable that you want to predict. Briefly explain what the target represents.

Your tasks

1. Explore and understand the dataset

- Load the dataset and inspect its shape, columns, data types, and basic descriptive statistics.
- Check for missing values, duplicated observations, unusual values, and obvious data-quality problems.
- Create a few relevant visualizations to show important relationships between variables and the target.
- Briefly describe the main observations from your exploration.

2. Prepare the data

- Select the features that will be used as model inputs and define the target variable.
- Handle missing values and categorical variables when necessary.
- Apply feature scaling when it is appropriate for the algorithms you choose.
- Explain the preprocessing decisions you make.

3. Create training and test data

Split the dataset into training and test sets before fitting the models. A common choice is 80% for training and 20% for testing. Keep the test set separate until the final evaluation.

Validation for model selection and tuning

Use 5-fold cross-validation on the training data when comparing model settings or tuning hyperparameters. This means you do not need a separate fixed validation set for this lab. The test set should only be used after model selection and tuning are completed.

4. Build at least two regression models

Train at least two different regression algorithms using the same training data. Suitable examples include Linear Regression, Decision Tree Regression, Random Forest Regression, K-Nearest Neighbors Regression, Support Vector Regression, or another regression method discussed in the course.

5. Evaluate the models

- Evaluate both models on the test set using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R^2).
- Present the results clearly and explain what the metrics indicate about model performance.
- Use the same test set for both models so that the comparison is fair.

6. Improve and tune the models

- Choose meaningful hyperparameters for each model and investigate whether tuning improves performance.
- Use cross-validation on the training data during tuning.
- Compare the performance before and after tuning and briefly discuss the result.

7. Interpret and compare the results

- Investigate which features appear most influential for at least one of your models, using an appropriate method.
- Compare the strengths and weaknesses of the two models.
- State which model you recommend for the selected dataset and justify your choice using the results.

Submission

Submit the following two files through Canvas:

- One clearly documented notebook containing your complete workflow and code. You may work in Jupyter Notebook, VS Code, Google Colab, or another environment of your choice. The notebook should include data preparation, model training, evaluation, tuning, and results. Use Markdown/text cells and meaningful comments so that your work is easy to follow.
- One PDF report with your discussion and final conclusions. The report should focus on interpreting and comparing the models rather than repeating the code.

In the PDF report, answer the following points:

- Compare the performance of the two models using MAE, RMSE, and R-squared (R^2), and explain what the results mean for your prediction problem.
- Discuss the most influential features identified using an appropriate method.
- Compare the strengths and weaknesses of the two models.
- State which model you recommend for the selected dataset and justify your choice using the obtained results.